

// Print all the nodes reachable from a given starting node in a digraph using BFS method.

```
import java.util.*;
```

```
public class P8_ReachableNodes {
```

```
    private int vertices;
```

```
    private ArrayList<LinkedList<Integer>> adjList;
```

```
    P8_ReachableNodes(int v) {
```

```
        vertices = v;
```

```
        adjList = new ArrayList<>(v);
```

```
        for (int i = 0; i < v; i++) {
```

```
            adjList.add(new LinkedList<>());
```

```
        }
```

```
    }
```

```
    void addEdge(int src, int dest) {
```

```
        adjList.get(src).add(dest);
```

```
    }
```

```
    void BFS(int start) {
```

```
        boolean[] visited = new boolean[vertices];
```

```
        Queue<Integer> queue = new LinkedList<>();
```

```
        visited[start] = true;
```

```
        queue.add(start);
```

```
        System.out.println("Nodes reachable from node " + start + " :");
```

```
        while (!queue.isEmpty()) {
```

```
            int node = queue.poll();
```

```
            System.out.print(node + " ");
```

```

        for (int neighbor : adjList.get(node)) {
            if (!visited[neighbor]) {
                visited[neighbor] = true;
                queue.add(neighbor);
            }
        }
    }
    System.out.println();
}

```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number of vertices: ");
    int v = sc.nextInt();

    P8_ReachableNodes graph = new P8_ReachableNodes(v);

    System.out.print("Enter number of edges: ");
    int e = sc.nextInt();

    System.out.println("Enter edges (src dest):");
    for (int i = 0; i < e; i++) {
        int src = sc.nextInt();
        int dest = sc.nextInt();
        graph.addEdge(src, dest);
    }

    System.out.print("Enter starting node: ");
    int start = sc.nextInt();
}

```

```
graph.BFS(start);  
sc.close();  
}  
}
```

```
Enter number of vertices: 6  
Enter number of edges: 6  
Enter edges (src dest):  
0  
1  
0  
2  
1  
3  
2  
4  
4  
  
5  
3  
5  
Enter starting node: 0  
Nodes reachable from node 0:  
0 1 2 3 4 5
```